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Enhancing Cybersecurity Visibility of OT Assets in Critical Infrastructure: A Practical Approach

Presented By

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INTRODUCTION

- Operational Technology (OT) powers critical infrastructure such as power grids and utilities
- Accurate OT asset inventory is essential for **cybersecurity, maintenance, and compliance**
- Traditional methods rely on

  spreadsheets → fragmented, error-prone, hard to scale

Existing Approaches & Limitations

- Existing Approaches
  - Spreadsheet-based inventory
    - Manual, inconsistent, difficult to maintain
  - Automated OT discovery tools
    - Not suitable for legacy/air-gapped networks
- Therefore, need a simple, low-risk, OT-friendly inventory system

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METHODOLOGY

We propose a **centralized OT Asset Inventory Management System**

- ❑ Define Asset Categories
  - IEDs, PLCs, RTUs, HMIs, Gateways, Switches
- ❑ Standardized Data Templates
  - Make, model, serial number
  - Firmware version
  - Network parameters
- ❑ Field Data Collection
  - Regional engineers & technicians
- ❑ Central Upload & Validation
  - Duplicate detection
  - IP format validation

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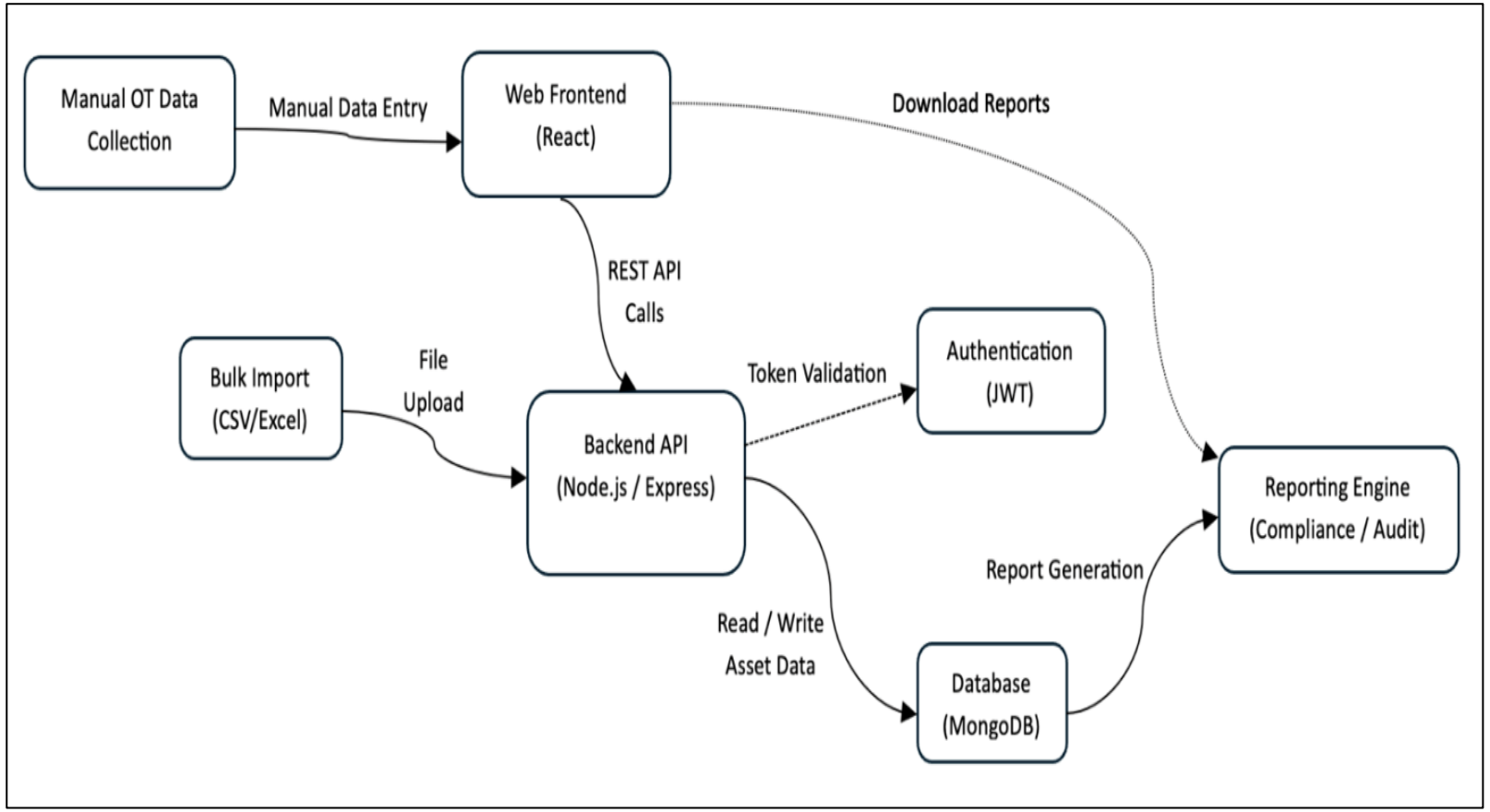
SYSTEM IMPLEMENTATION

- Technology Stack
  - Frontend – React UI
  - Backend
    - Node.js
    - Express API
  - Database
    - MongoDB
  - Security
    - JWT Authentication
    - bcrypt password hashing
- Key Capabilities – REST API integration, Asset reporting & export



PILOT DEPLOYMENT AND RESULTS

- Pilot deployment was done in a lab environment
- Evaluated against the metrics such as accuracy of asset records, reduction in manual errors, ease of use, etc.
- Key Results
  - Improved visibility of OT asset information in the centralized system
  - Reduction in manual errors and data redundancy
  - Enhanced compliance and audit readiness
  - Improved decision-making for OT Cybersecurity

CHALLENGES AND DISCUSSION

- Legacy OT devices, data standardization
- Diverse naming conventions across substations
- Limited use of automated scanning tools due to operational risk
- Major challenge is maintaining inventory accuracy over time
- Therefore, a model involving asset ownership roles, periodic review & validation is required.

CONCLUSION AND FUTURE WORK

- OT Asset Inventory Management System
  - Provides a practical, low-risk, scalable solution
  - Maintains an accurate, centralized, and audit ready OT asset inventory
  - Addresses limitations of spreadsheet-based methods
  - Avoids operational risks associated with agent-based or automated scanning tools
- Future enhancements include field validation in operational substations, partial automation via passive monitoring interfaces, advanced analytics dashboards, etc.

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THANK YOU